

PRACTICAL – 11

Code:

```
GNU nano 7.2 race_condition.c
#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 4
#define ITERATIONS 100000

int counter = 0;

void *increment(void *arg)
{
    for (int i = 0; i < ITERATIONS; i++)
    {
        counter++;
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_create(&threads[i], NULL, increment, NULL);
    }
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ gcc race_condition.c -o race_condition -pthread
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ./race_condition
Expected counter value: 400000
Actual counter value: 202845
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ./race_condition
Expected counter value: 400000
Actual counter value: 277869
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ./race_condition
Expected counter value: 400000
Actual counter value: 143899
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ^C
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ./race_condition
Expected counter value: 400000
Actual counter value: 205779
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ |
```

Part-C:

Code:

```
GNU nano 7.2 mutex_counter.c
#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 4
#define ITERATIONS 100000

int counter = 0;
pthread_mutex_t lock;

void *increment(void *arg)
{
    for (int i = 0; i < ITERATIONS; i++)
    {
        pthread_mutex_lock(&lock);

        counter++;

        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main()
{
}
```

Output:

```

Expected counter value: 400000
Actual counter value: 400000
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ./mutex_counter
Expected counter value: 400000
Actual counter value: 400000
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ./mutex_counter
Expected counter value: 400000
Actual counter value: 400000
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ./mutex_counter
Expected counter value: 400000
Actual counter value: 400000
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ./mutex_counter
Expected counter value: 400000
Actual counter value: 400000
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ |

```

PART D — Check Files

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ls -l
total 48
-rw-r--r-- 1 bhardwaj bhardwaj 1 Oct 1 09:43 hello
-rwxr-xr-x 1 bhardwaj bhardwaj 16440 Oct 1 09:48 mutex_counter
-rw-r--r-- 1 bhardwaj bhardwaj 810 Oct 1 09:48 mutex_counter.c
-rwxr-xr-x 1 bhardwaj bhardwaj 16184 Oct 1 09:45 race_condition
-rw-r--r-- 1 bhardwaj bhardwaj 640 Oct 1 09:44 race_condition.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ rm race_condition mutex_counter
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ ls -l
total 12
-rw-r--r-- 1 bhardwaj bhardwaj 1 Oct 1 09:43 hello
-rw-r--r-- 1 bhardwaj bhardwaj 810 Oct 1 09:48 mutex_counter.c
-rw-r--r-- 1 bhardwaj bhardwaj 640 Oct 1 09:44 race_condition.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-11$ |

```